

6406532879016. ✖ 0x8000000000000009

6406532879017. ✔ 0x8000000000000005

6406532879018. ✖ 0x0000000000000001

6406532879019. ✖ 0x0000000000000008

Game Theory

Section Id :	64065361125
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	8
Number of Questions to be attempted :	8
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653127975
Question Shuffling Allowed :	No

Question Number : 50 Question Id : 640653856808 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS

REGISTERED BY YOU)

Options :

6406532879028. ✓ YES

6406532879029. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653127976

Question Shuffling Allowed :

No

Question Id : 640653856809 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (51 to 57)

Question Label : Comprehension

Consider the example of cascade discussed in the lecture but with a modification that now the urn is containing either 3 blue balls and 1 red ball or 3 red balls and 1 blue ball with equal probability. That simply means that the urn is equally likely to be blue-majority urn or red-majority urn. We assume everyone hears what the previous individuals report and get their own private signal by randomly drawing one ball from the urn.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 51 Question Id : 640653856810 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Suppose one ball is picked from the red majority urn. What is the probability that the ball is blue?

Options :

6406532879030. ✓ 1/4

6406532879031. ✗ 3/4

6406532879032. ✗ 1/5

6406532879033. ✗ 3/5

Question Number : 52 Question Id : 640653856811 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the probability that the urn is red-majority if the first student draws a red ball?

Options :

6406532879034. ✗ 2/5

6406532879035. ✗ 3/5

6406532879036. ✓ 3/4

6406532879037. ✗ 1/4

Question Number : 53 Question Id : 640653856812 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Should the first student guess the urn to be blue majority (in continuation to the previous question)?

Options :

6406532879038. ✖ Yes

6406532879039. ✔ No

Question Number : 54 Question Id : 640653856813 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Red, Red, Blue).

What is the probability of getting this sequence if the urn is a blue-majority urn?

Options :

6406532879040. ✔ $3/64$

6406532879041. ✖ $9/64$

6406532879042. ✖ $3/16$

6406532879043. ✖ $9/16$

Question Number : 55 Question Id : 640653856814 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Red, Red, Blue).

What is the probability of getting this sequence if the urn is a red-majority urn?

Options :

6406532879044. ✖ $3/64$

6406532879045. ✔ $9/64$

6406532879046. ✖ $3/16$

6406532879047. ✖ $9/16$

Question Number : 56 Question Id : 640653856815 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Red, Red, Blue).

What is the total probability of getting this sequence?

Options :

6406532879048. ✖ $3/16$

6406532879049. ✖ $9/32$

6406532879050. ✖ $9/16$

6406532879051. ✓ 3/32

Question Number : 57 Question Id : 640653856816 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Now suppose, first three students draw the ball in the sequence (Red, Red, Blue).

What is the probability of urn being red-majority in this scenario?

Options :

6406532879052. ✗ 3/16

6406532879053. ✗ 1/4

6406532879054. ✗ 3/64

6406532879055. ✓ 3/4

Sub-Section Number :

3

Sub-Section Id :

640653127977

Question Shuffling Allowed :

Yes

Question Number : 58 Question Id : 640653856817 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Three sweets are running in an election for the most popular sweet: Gulab jamun (G), Malpua (M), and Balushahi (B). The preference schedule is shown below

	No. of votes in thousands						
Rankings	44	14	20	70	22	80	39
I Choice	G	G	G	M	M	B	B
II Choice	M	B	-	G	B	M	-
III Choice	B	M	-	B	G	G	-

Which candidate wins under the plurality method?

Options :

6406532879056. ✗ M

6406532879057. ✓ B

6406532879058. ✗ G

6406532879059. ✗ Cannot be determined

Sub-Section Number :

4

Sub-Section Id :

640653127978

Question Shuffling Allowed :

No

Question Id : 640653856818 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (59 to 60)

Question Label : Comprehension

Suppose there are four alternatives, named A, B, C, and D. There are three voters who have the following individual rankings:

$$B \succ_1 C \succ_1 D \succ_1 A$$

$$C \succ_2 D \succ_2 A \succ_2 B$$

$$D \succ_3 A \succ_3 B \succ_3 C$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 59 Question Id : 640653856819 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Using Borda count, the group ranking will be

Options :

6406532879060. ✖ $B \succ C \succ D \succ A$

6406532879061. ✔ $D \succ C \succ B \succ A$

6406532879062. ✖ $C \succ D \succ A \succ B$

6406532879063. ✖ $D \succ A \succ B \succ C$

Question Number : 60 Question Id : 640653856820 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Consider two statements about Borda count:

I: Borda Count (ignoring ties) always produces a complete, transitive ranking for a set of alternatives.

II: Outcome in Borda count depends on the presence of alternatives that seem “irrelevant”.

Choose the correct alternative about these statements:

Options :

6406532879064. ✖ Only I is true

6406532879065. ✖ Only II is true

6406532879066. ✖ None is true

6406532879067. ✔ Both I and II are true

Sub-Section Number :

5

Sub-Section Id :

640653127979

Question Shuffling Allowed :

No

Question Id : 640653856821 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (61 to 64)

Question Label : Comprehension

Find a stable matching system for the given preference structure using the Gale–Shapley algorithm, when the men propose to the women:

Women: A, B, C, D.

Men: a, b, c, d.

	A	B	C	D
a	1	2	4	2
b	2	4	2	1
c	3	1	1	3
d	4	3	3	4

	A	B	C	D
a	4	2	1	3
b	2	1	3	4
c	3	1	4	2
d	2	4	1	3

In a stable matching when men propose to women

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 61 Question Id : 640653856822 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

A will be matched with

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : Yes

Text Areas : PlainText

Possible Answers :

b

Question Number : 62 Question Id : 640653856823 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

B will be matched with

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : Yes

Text Areas : PlainText

Possible Answers :

c

Question Number : 63 Question Id : 640653856824 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

C will be matched with

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : Yes

Text Areas : PlainText

Possible Answers :

d

Question Number : 64 Question Id : 640653856825 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

D will be matched with

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : Yes

Text Areas : PlainText

Possible Answers :

a

Question Id : 640653856841 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (65 to 66)

Question Label : Comprehension

Consider the following prisoners' dilemma game and answer the given subquestions.

	Cooperate	Defect
Cooperate	2, 2	0, 3
Defect	3, 0	1, 1

Sub questions

Question Number : 65 Question Id : 640653856842 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Unique NE of this game is

Options :

6406532879100. ✓ (Defect, Defect)

6406532879101. ✗ (Cooperate, Cooperate)

6406532879102. ✗ (Defect, Defect) and (Cooperate, Cooperate)

6406532879103. ✗ None of these

Question Number : 66 Question Id : 640653856843 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Suppose, this game is repeatedly played finite times

Options :

6406532879104. ✗ In the Last Period Cooperate is a dominant strategy irrespective of history of the game

6406532879105. ✓ In the Last Period Defect is a dominant strategy irrespective of history of the game

6406532879106. ✗ In the Last Period Cooperate is a dominant strategy for a particular game history

6406532879107. ✗ In the Last Period Defect is a dominant strategy for a particular game history

Sub-Section Number :

6

Sub-Section Id :

640653127980

Question Shuffling Allowed :

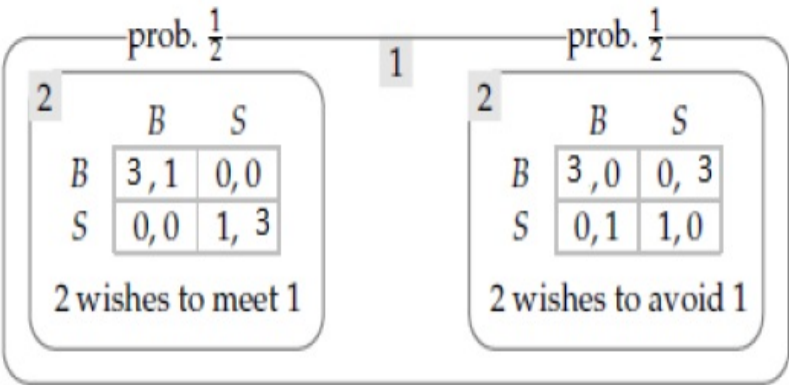
No

Question Id : 640653856826 Question Type : COMPREHENSION Sub Question Shuffling
Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (67 to 75)

Question Label : Comprehension

(Bayesian Game) Consider a variant of the situation modelled by *Battle of Sexes (BoS)* (as discussed in lectures) in which player 1 is unsure whether player 2 prefers to go out with her or prefers to avoid her, whereas player 2, as before, knows player 1's preferences. Specifically, suppose player 1 thinks that with probability $1/2$ player 2 wants to go out with her, and with probability $1/2$ player 2 wants to avoid her. The situation is depicted in the following matrix of a static game with incomplete information:



With the help of this game matrix, we may come up with the following table depicting expected payoff of player 1 corresponding to different actions of both types of player 2.

		Player 2 (both types)			
		(B, B)	(B, S)	(S, B)	(S, S)
Player 1	B	a	b	c	d
	S	p	q	r	t

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 67 Question Id : 640653856827 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of a =

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

Question Number : 68 Question Id : 640653856828 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $b = \dots\dots\dots$

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1.5

Question Number : 69 Question Id : 640653856829 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $c = \dots\dots\dots$

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1.5

Question Number : 70 Question Id : 640653856830 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $d = \dots\dots\dots$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 71 Question Id : 640653856831 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $p = \dots\dots\dots$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Question Number : 72 Question Id : 640653856832 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $q = \dots\dots\dots$

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Question Number : 73 Question Id : 640653856833 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $r = \dots\dots\dots$

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Question Number : 74 Question Id : 640653856834 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

Find out the value of $t = \dots\dots\dots$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 75 Question Id : 640653856835 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What is the Bayesian Nash equilibrium in this game?

Options :

6406532879080. ✖ (S, (S, B))

6406532879081. ✔ (B, (B, S))

6406532879082. ✖ (S, (B, S))

6406532879083. ✖ (B, (B, B))

Sub-Section Number :

7

Sub-Section Id :

640653127981

Question Shuffling Allowed :

No

Question Id : 640653856836 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (76 to 79)

Question Label : Comprehension

Consider the following strategic games and answer the given subquestions.

Table 1: Game 1

	X	Y
X	3, 3	0, 0
Y	0, 0	0, 0

Table 2: Game 2

	X	Y
X	3, 3	0, 0
Y	0, 0	2, 2

Sub questions

Question Number : 76 Question Id : 640653856837 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

NE of the Game 1 is (are)

Options :

6406532879084. ✖ (X, X)

6406532879085. ✖ (Y, Y)

6406532879086. ✔ (X, X) and (Y,Y)

6406532879087. ✖ None of these

Question Number : 77 Question Id : 640653856838 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

NE of the Game 2 is (are)

Options :

6406532879088. ✖ (X, X)

6406532879089. ✖ (Y, Y)

6406532879090. ✔ (X, X) and (Y,Y)

6406532879091. ✖ None of these

Question Number : 78 Question Id : 640653856839 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Evolutionary Stable Strategy (ESS) in Game 1 is (are)

Options :

6406532879092. ✔ X